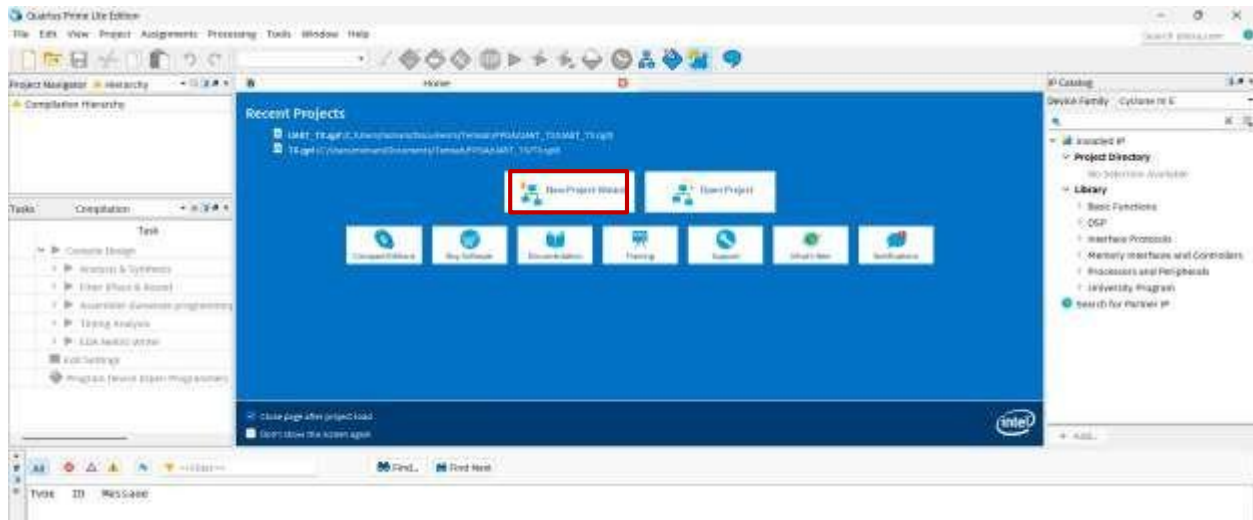


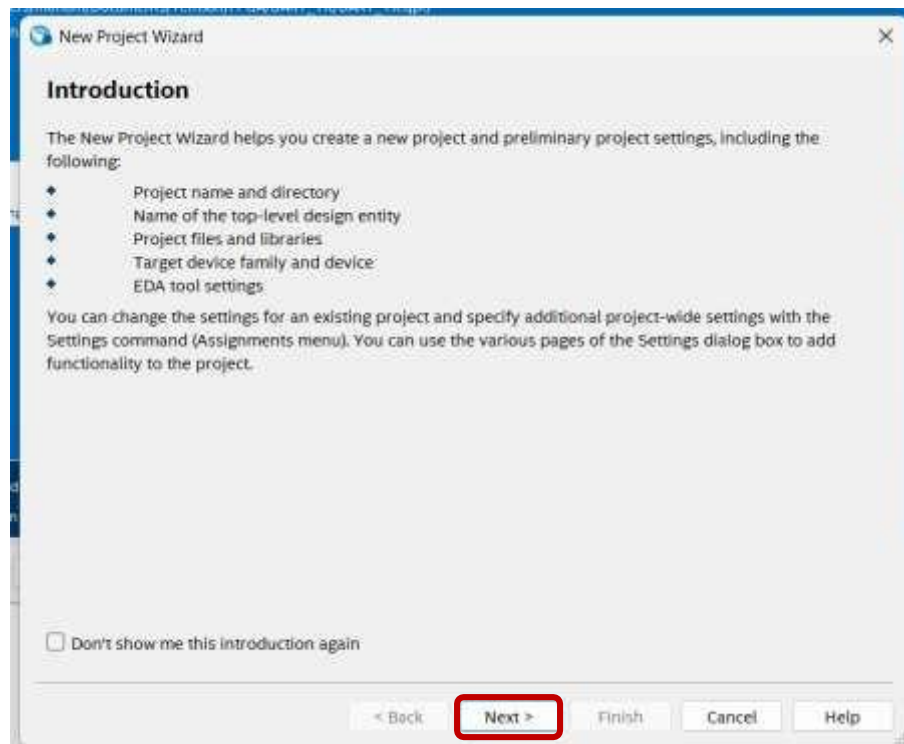
Lab2

Step 1: Project setup

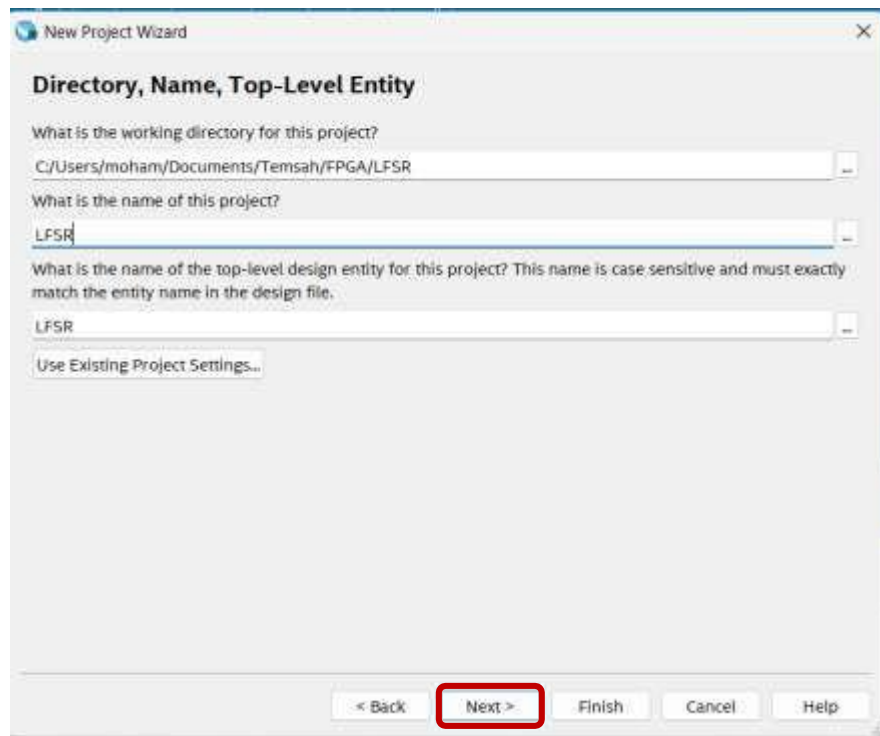
- Launch **Intel Quartus Prime Lite Edition** and click on **New Project Wizard** on the home landing page.
- Initializes a new design workspace, where the project directory, top-level entity name (**top_module**), target FPGA family (**MAX 10** or **Cyclone IV**), and EDA tools are defined.



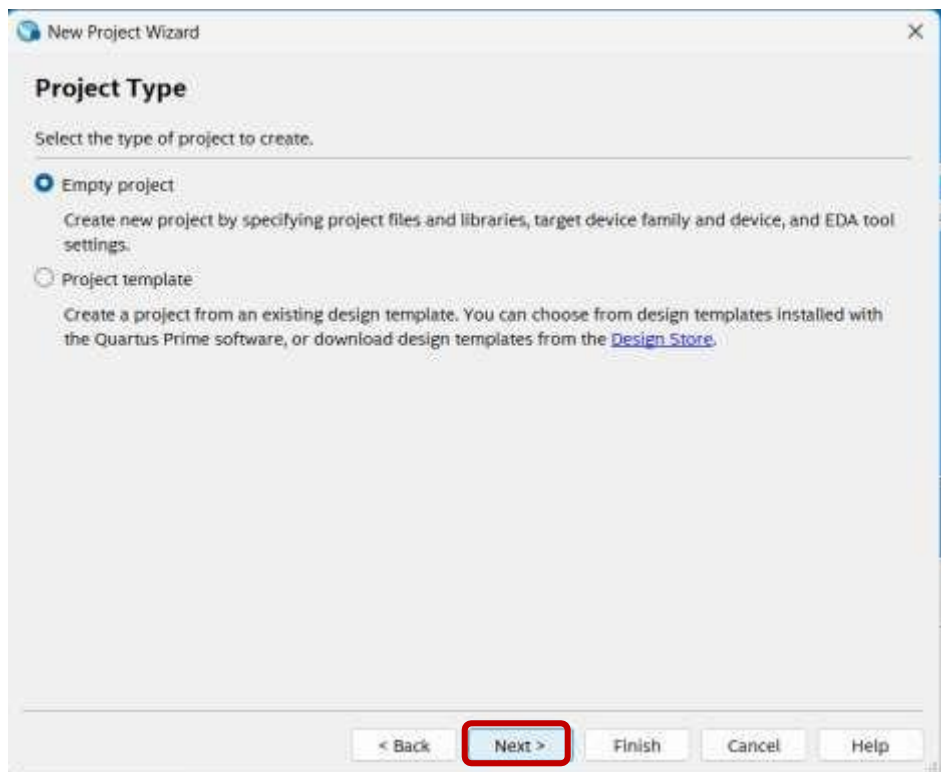
- Click **Next >** on the wizard introduction dialog to proceed with project configuration.



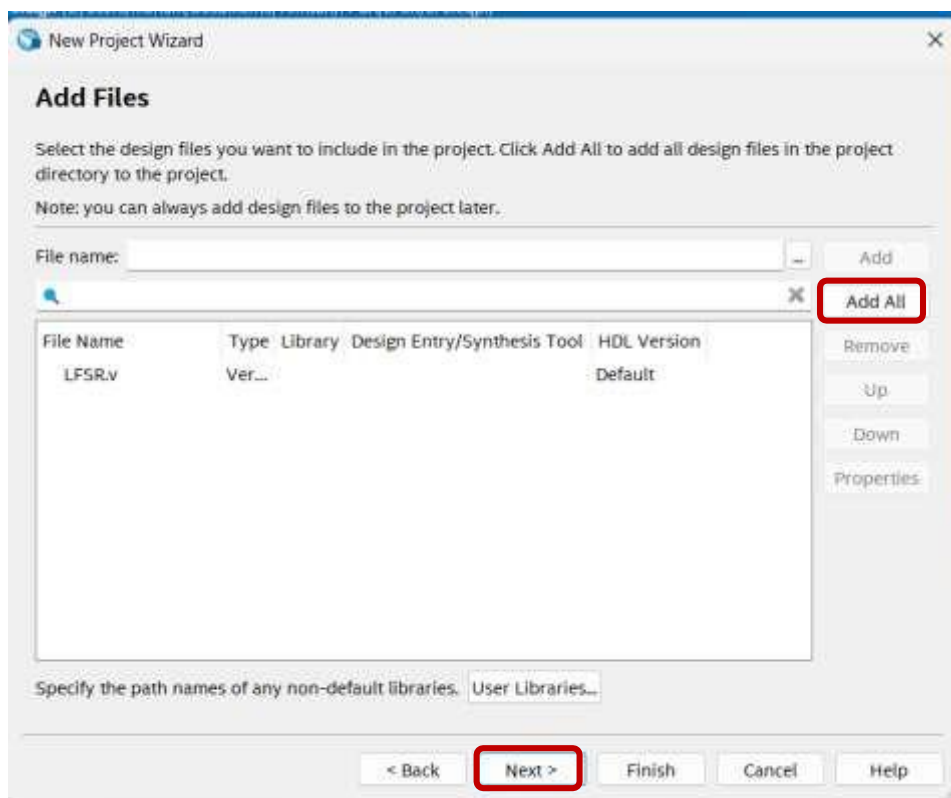
- Set the working directory path, set the project name to **LFSR**, and set the top-level entity name to **LFSR**. Click **Next >**.



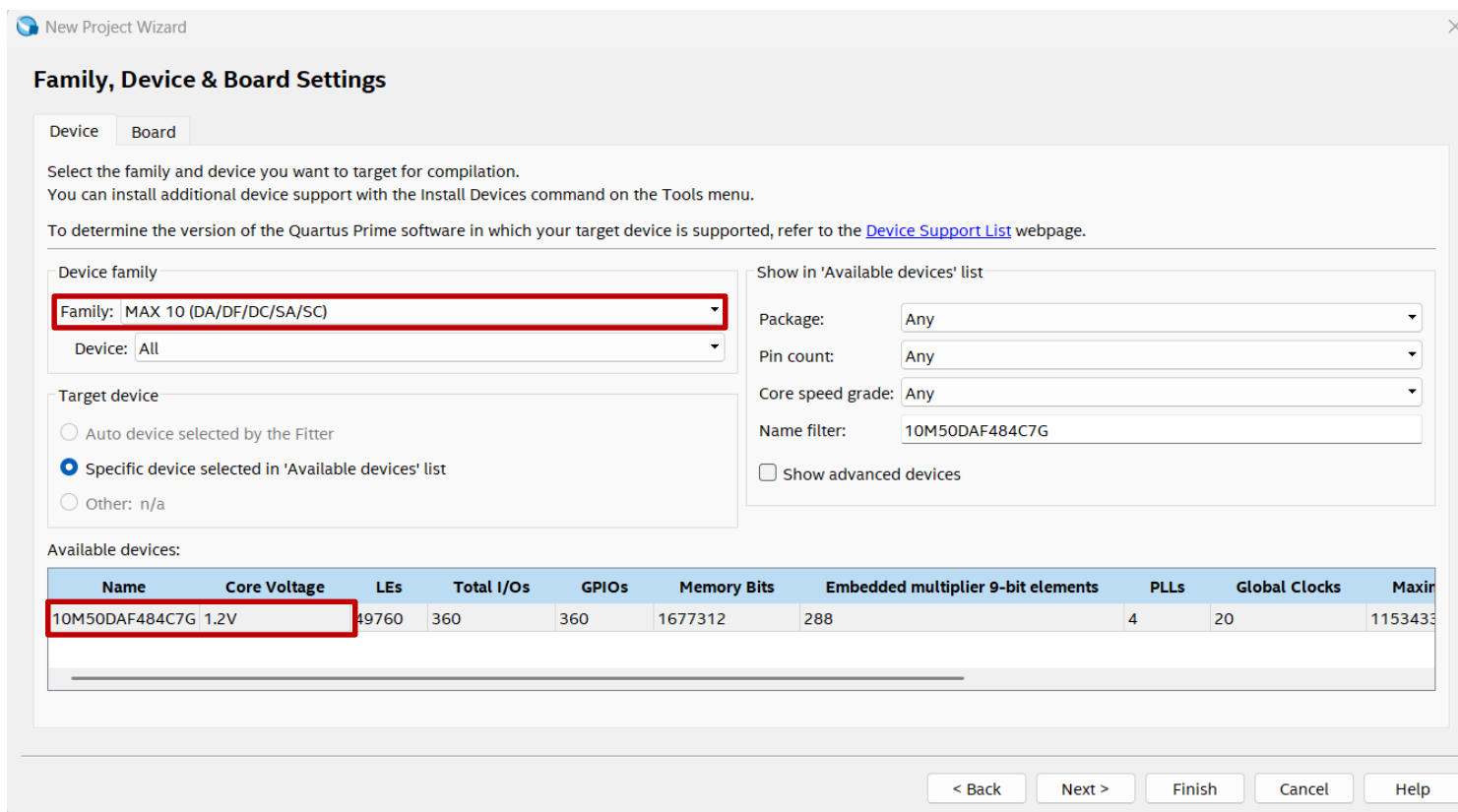
- Select **Empty project** and click **Next >** to create a fresh workspace for importing HDL source files and IP blocks.



- Add all Verilog design files (such as **LFSR.v**) to the project by clicking **Add All**, then click **Next >**.

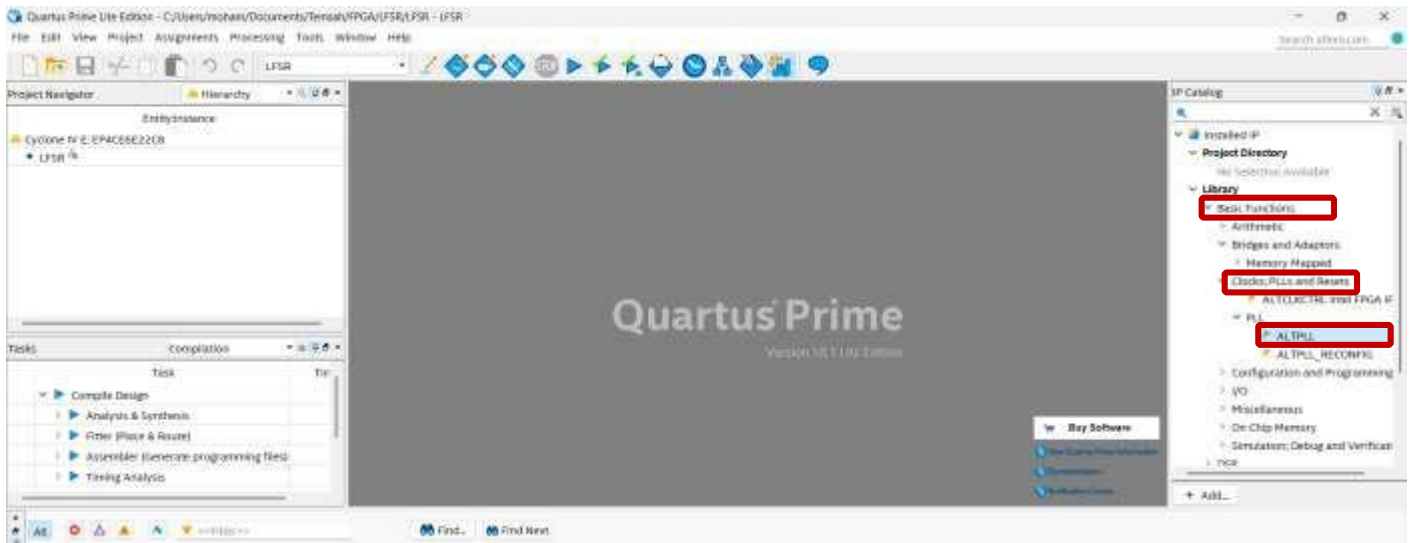


- Set the Device family to **MAX 10**, select the target FPGA device **10M50DAF484C7G**, and click **Finish** to instantiate the workspace.

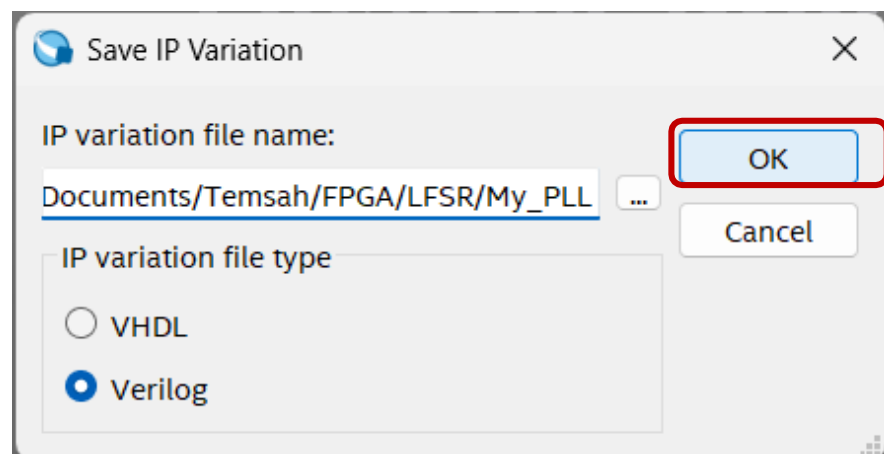


Step 2: Clock Management

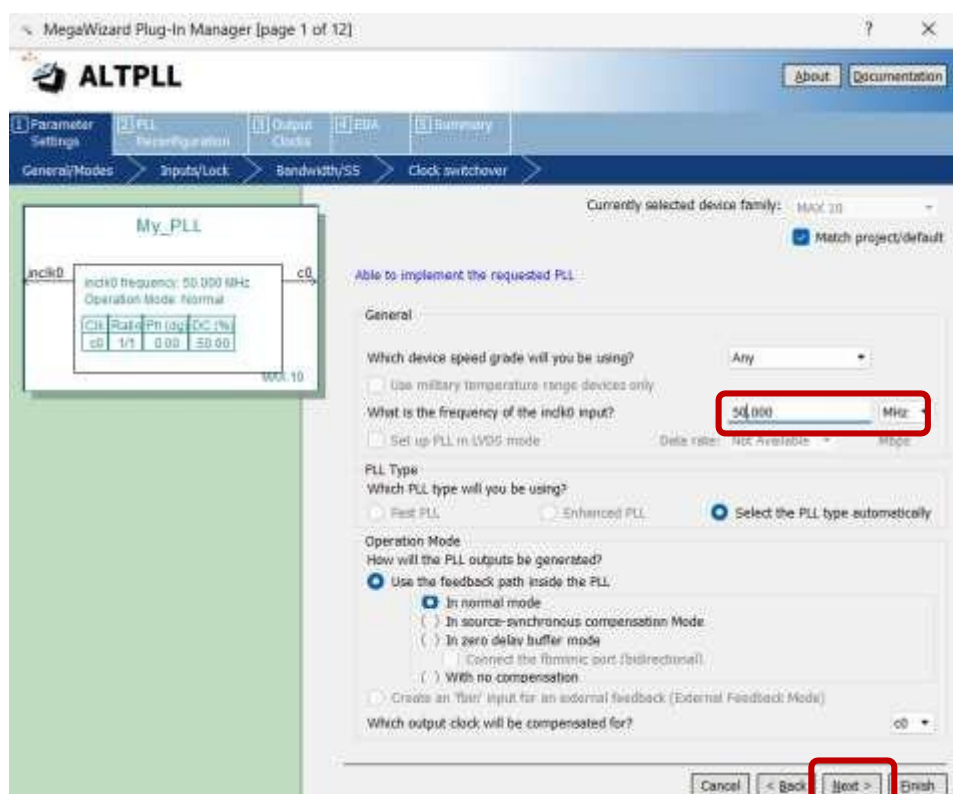
- In the **IP Catalog** panel, expand **Library** -> **Basic Functions** -> **Clocks, PLLs and Resets** -> **PLL**, and double-click **ALTPLL**.



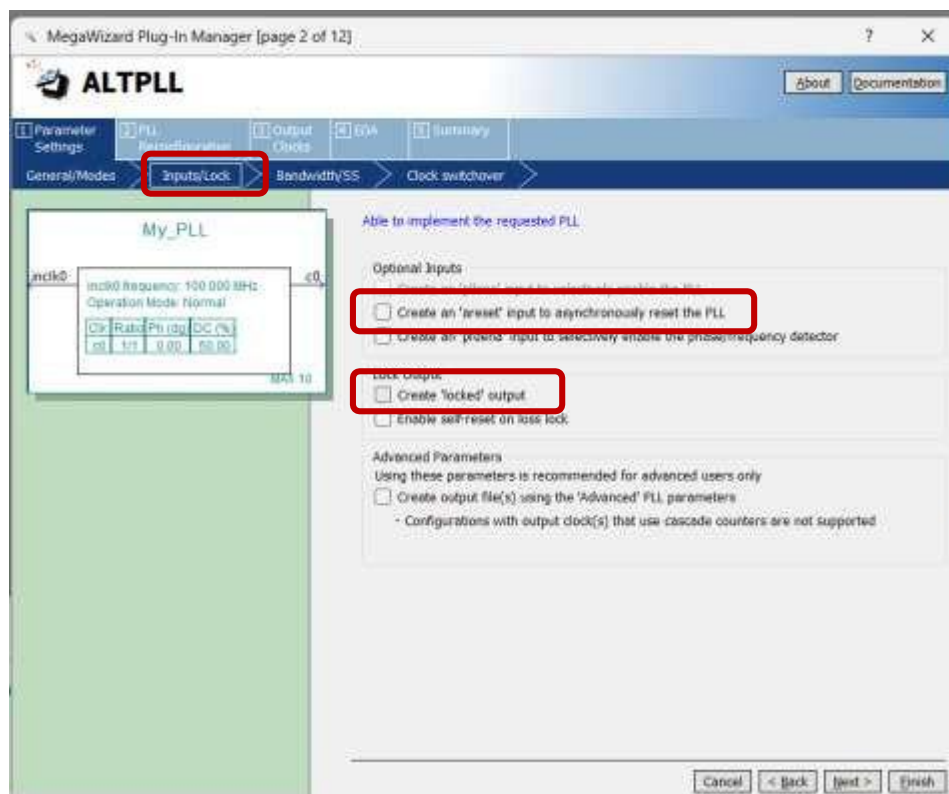
- Set the IP variation file name to **My_PLL**, select **Verilog** as the IP variation file type, and click **OK**.



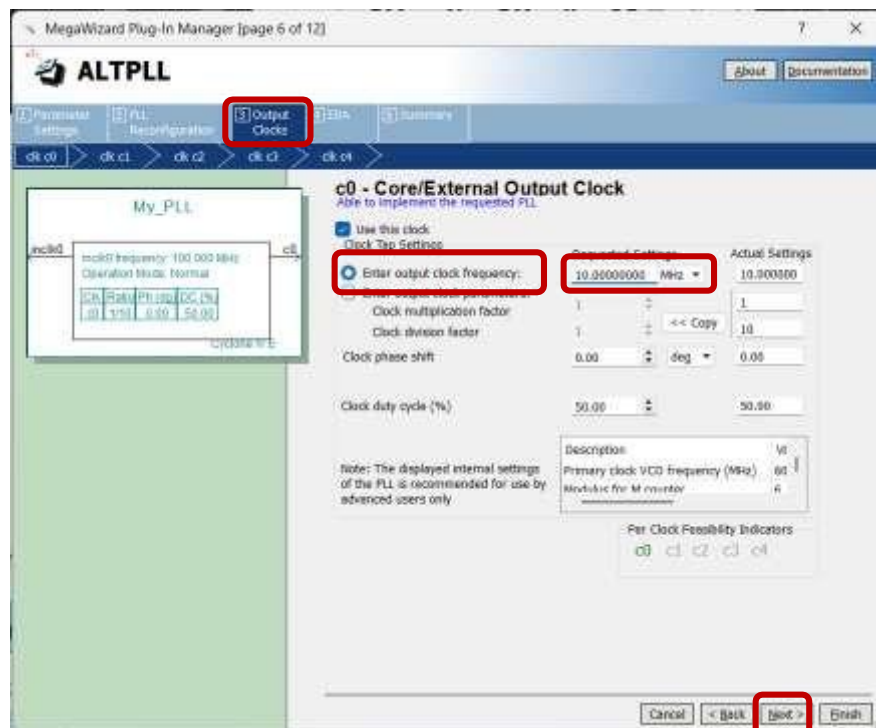
- On the **General/Modes** tab, set the inclk0 input clock frequency to **50.000 MHz** and click **Next >**.



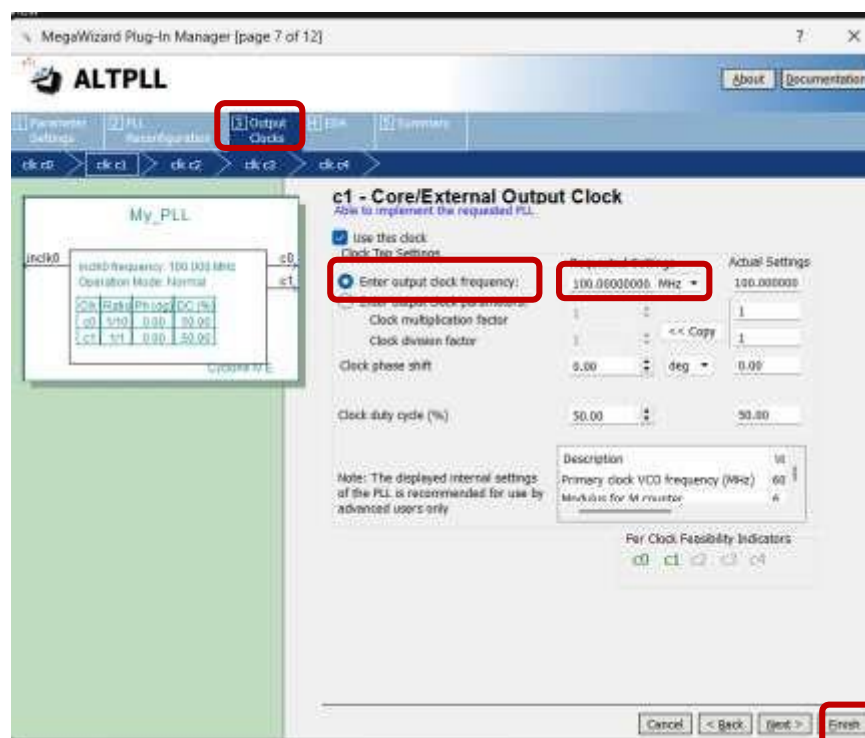
- On the **Inputs/Lock** tab, uncheck 'areset' (asynchronous reset) and 'locked' (lock output) to simplify the PLL clock interface, then click **Next >**.



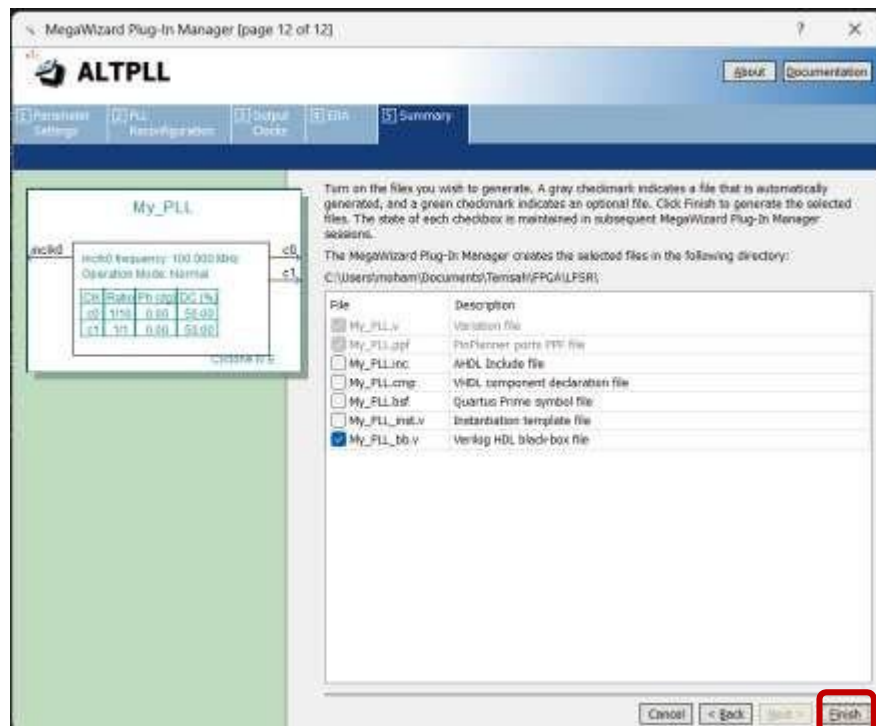
- On the **clk c0** tab, select **Enter output clock frequency** and set the frequency to **10.000 MHz** for system logic execution.



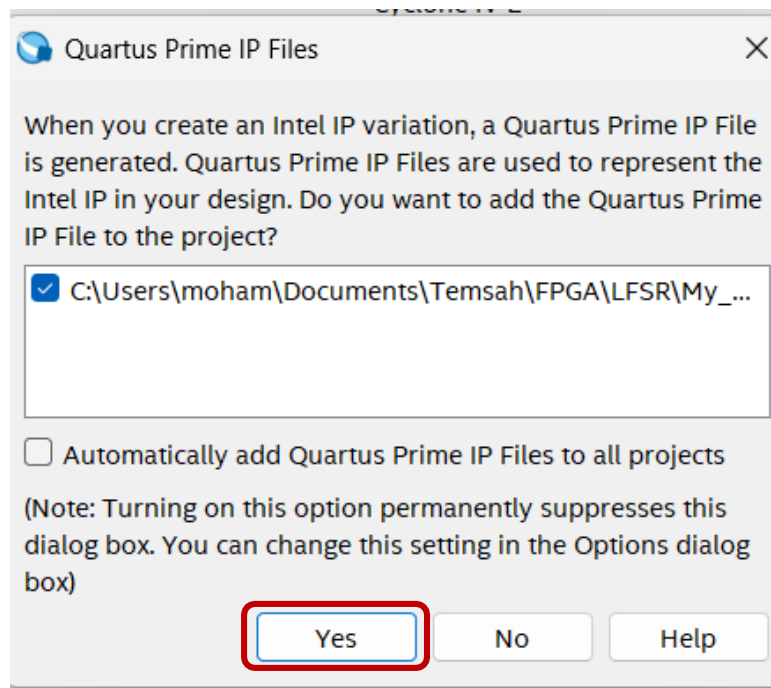
- On the **clk c1** tab, select **Enter output clock frequency** and set the frequency to **100.000 MHz** for Signal Tap oversampling.



- Navigate to the **Summary** tab, ensure the target IP output files are checked, and click **Finish** to generate the ALTPLL core.

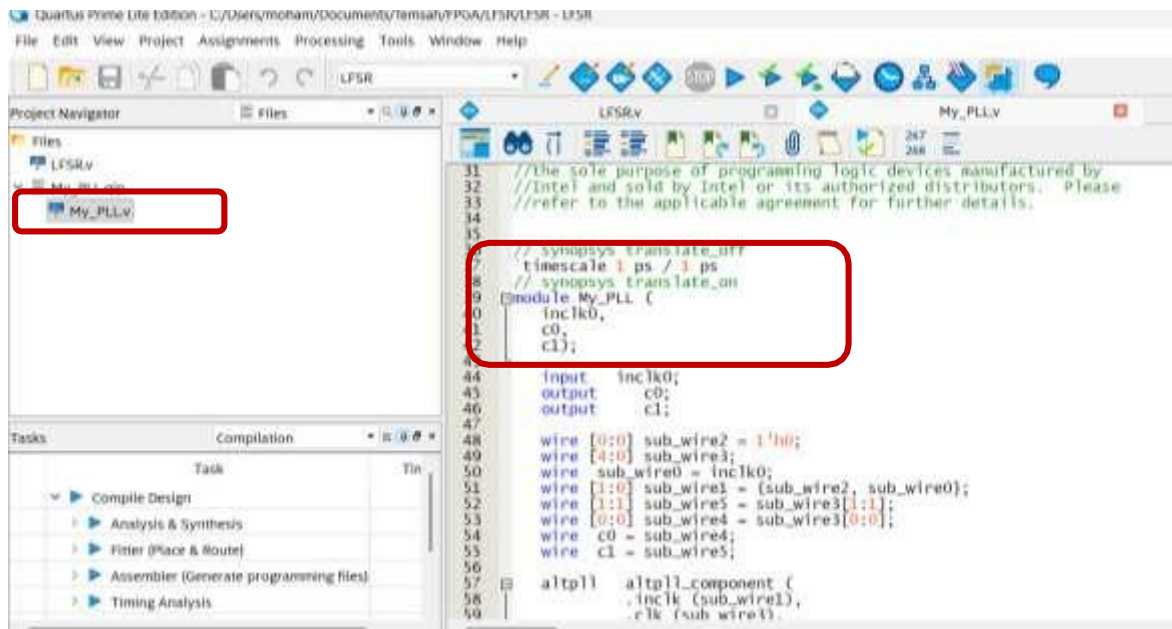


- Click **Yes** on the pop-up dialog to automatically add the generated IP variation file (**MY_PLL.qip**) to the Quartus project.



- Double-click **My_PLL.v** in the Project Navigator to inspect the generated module interface (incl0, c0, c1)

- Instantiate the **My_PLL** IP block inside the top-level wrapper (**top.v**) to route the 50 MHz input clock to the internal logic clock (c0) and Signal Tap sampling clock (c1).

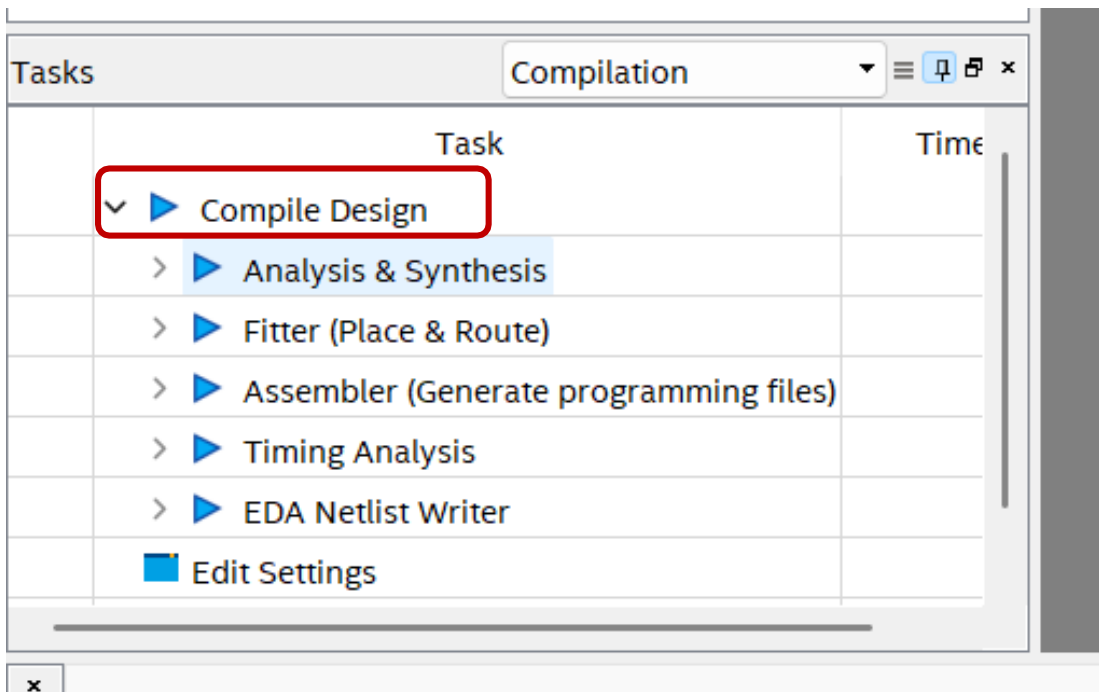
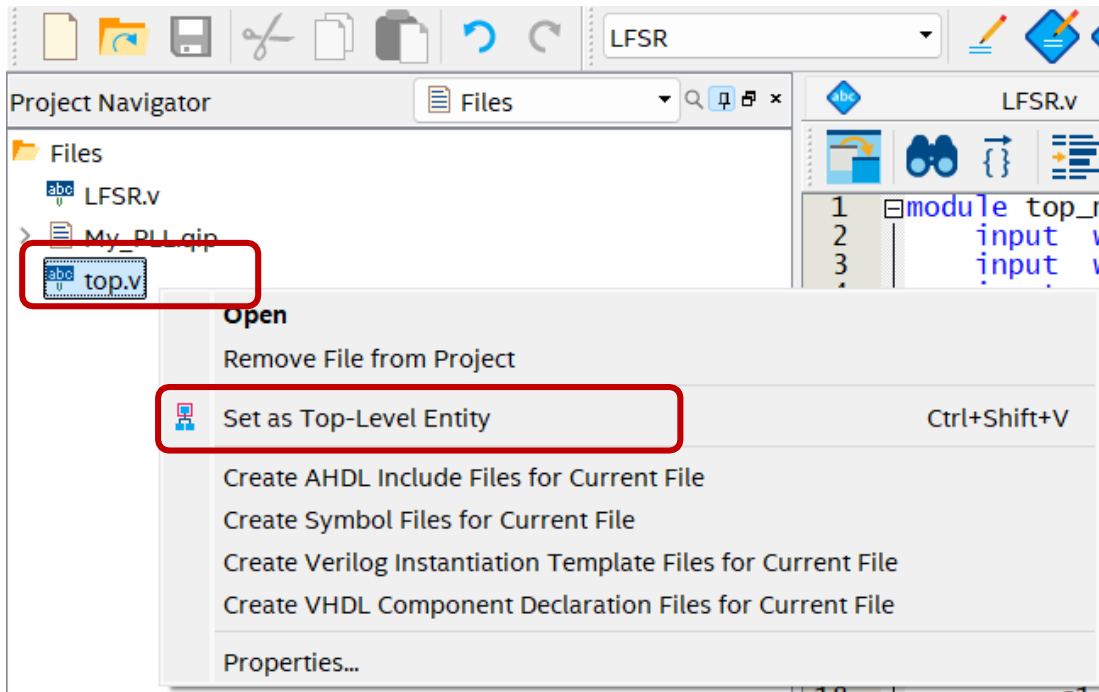


```

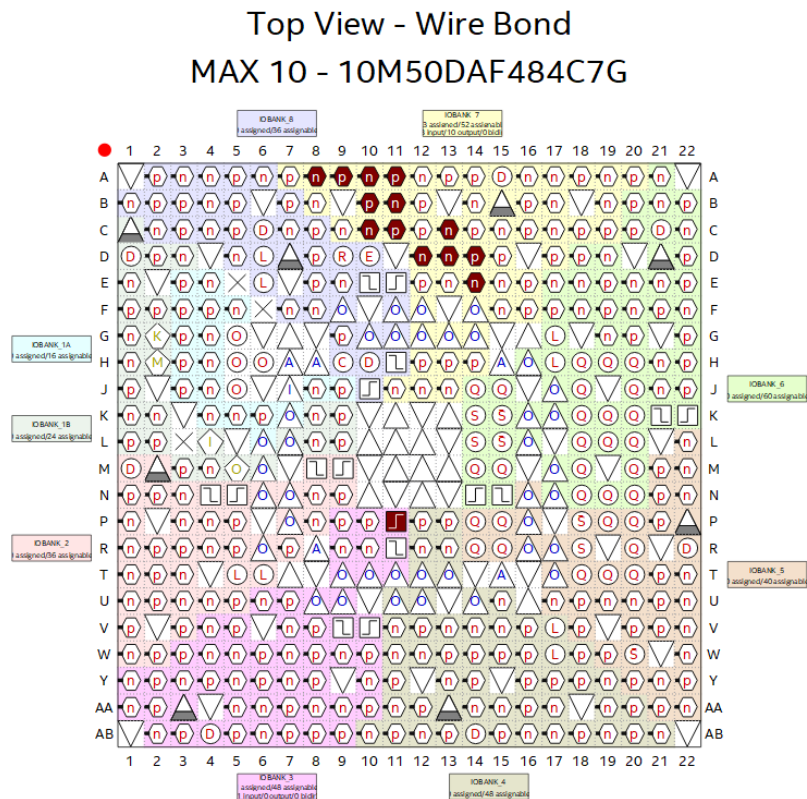
1 module top (
2     input wire      clk_50m, // PIN_23
3     input wire      rst_n,   // PIN_25
4     input wire      active_sw, // PIN_88
5     input wire      data_in, // PIN_89
6     output wire [7:0] leds,   // PINS 87, 86, 85, 84, 114, 115, 119, 120
7     output wire      crc_out, // PIN_112
8     output wire      valid_out, // PIN_111
9 );
10
11 wire clk_10m; // System logic clock (c0)
12 wire clk_100m; // Signal Tap 10x sampling clock (c1)
13
14 // ALTPLL Core Instance
15 My_PLL u_pll (
16     .inclk0 (clk_50m),
17     .c0      (clk_10m),
18     .c1      (clk_100m)
19 );
20
21 // LFSR Core Instance
22 LFSR u_crc (
23     .Clock      (clk_10m),
24     .Reset      (rst_n),
25     .Enable     (active_sw),
26     .OUT_Enable (~active_sw),
27     .Seed       (8'hA5), // Pre-loaded test seed
28     .OUT        (crc_out),
29     .valid      (valid_out),
30     .lfsr_state (leds)
31 );
32
33 endmodule

```

- Right-click **top.v** in the Project Navigator and select **Set as Top-Level Entity**.
- In the Tasks pane on the left, double-click **Analysis & Synthesis** to compile and verify the RTL design structure.

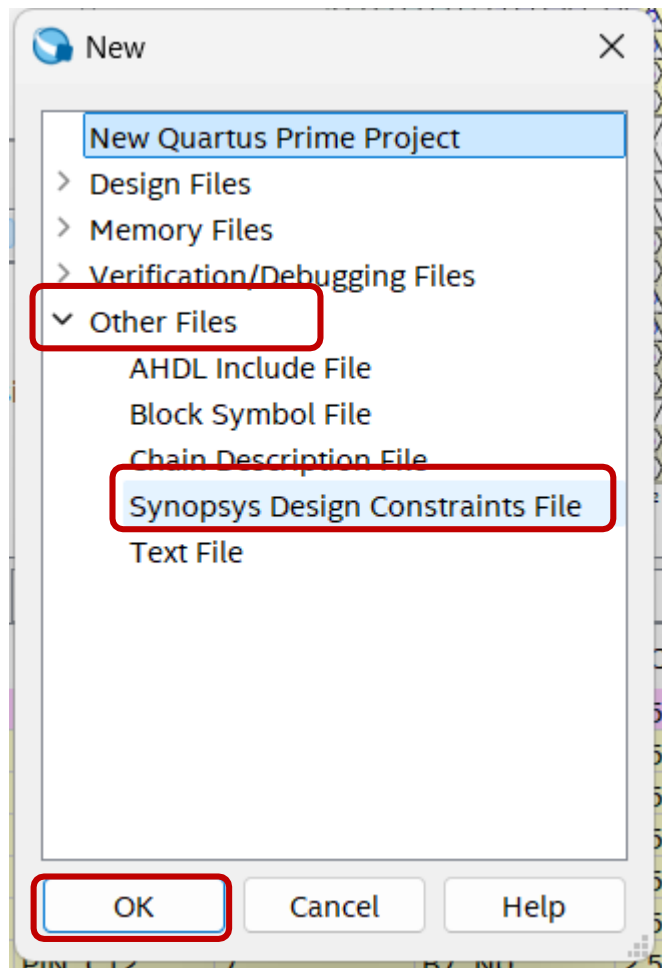


- Open **Pin Planner** by clicking the Pin Planner icon on the top toolbar (or **Ctrl + Shift + N**).
- Assign the physical target FPGA pin locations for clk_50m, rst_n, LFSR_EN, OUT_EN, LEDs[7:0], valid_out, and crc_out.

[illegible]

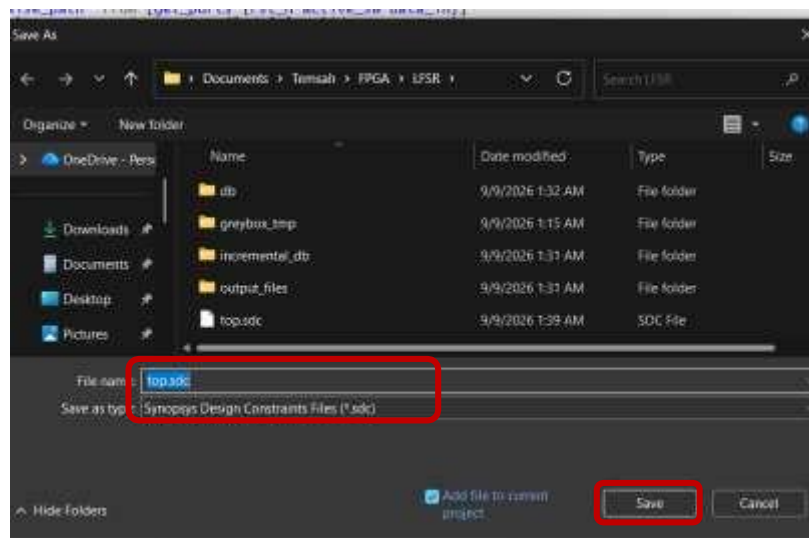
Step 4: Timing Constraints

- Navigate to File -> New, expand Other Files, select Synopsys Design Constraints File, and click OK.



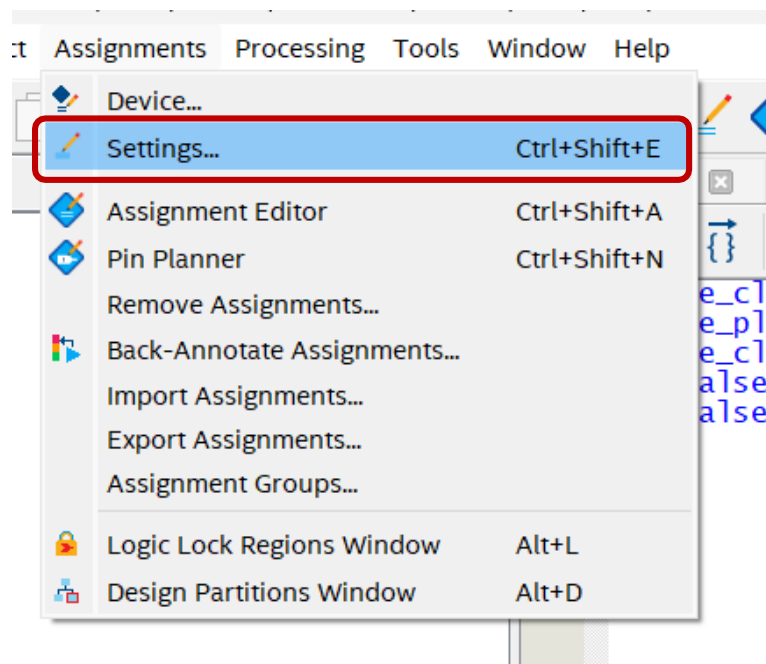
- Enter the SDC timing constraints (create_clock, derive_pll_clocks, set_false_path) into the editor.
- Save the constraint file as top.sdc in the root project directory.

```
create_clock -name clk_50m -period 20.000 [get_ports clk_50m]
derive_pll_clocks
derive_clock_uncertainty
set_input_delay 0.5 -clock clk_50m [get_ports {LFSR_EN OUT_EN data_in}]
set_output_delay 0.5 -clock clk_50m [get_ports {LEDs valid_out crc_out}]
```

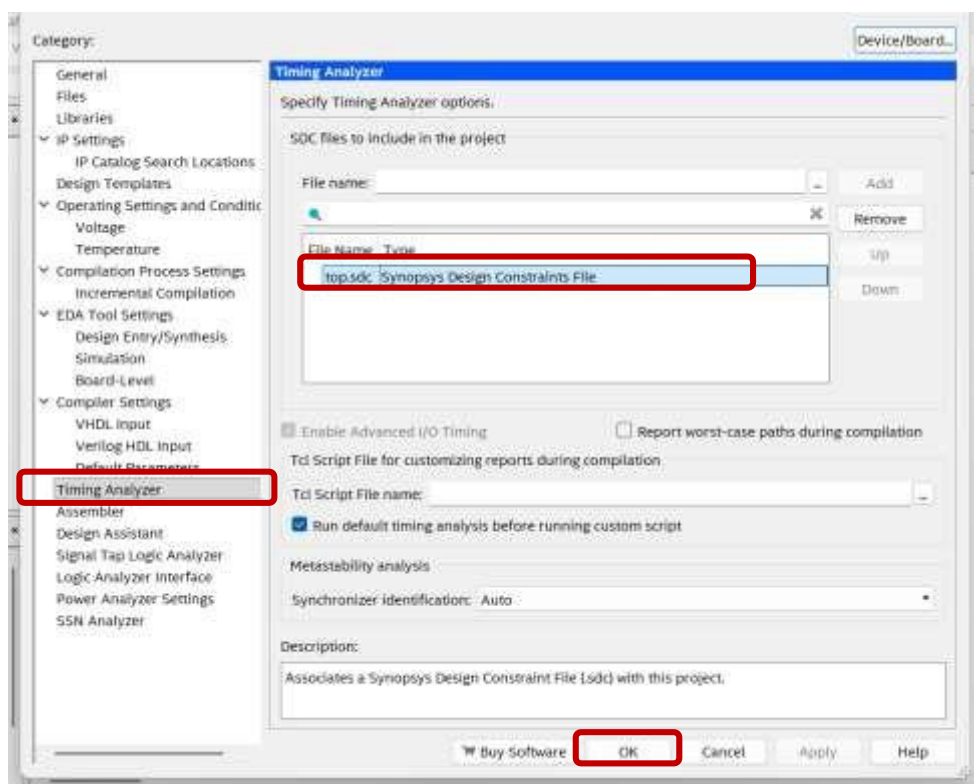


Step 5: Timing Analyzer

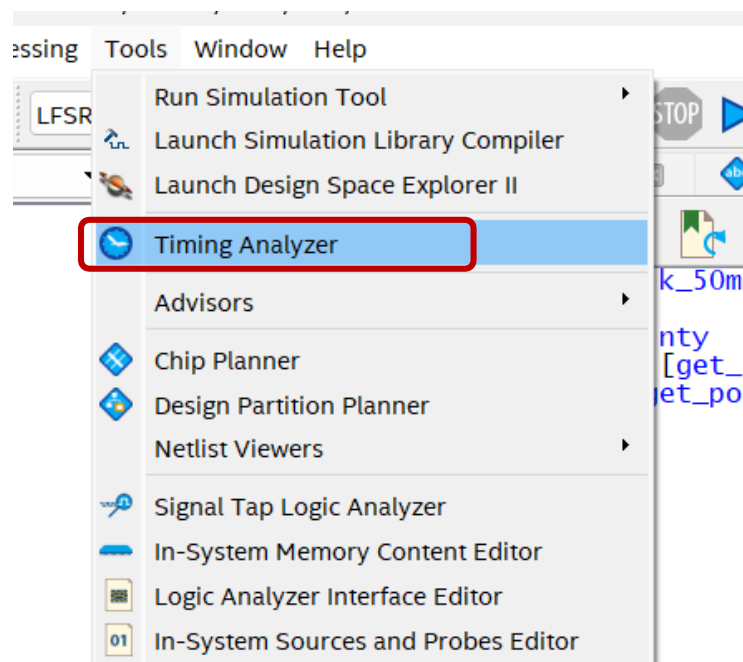
- Launch project settings by selecting **Assignments** -> **Settings** from the top menu bar.



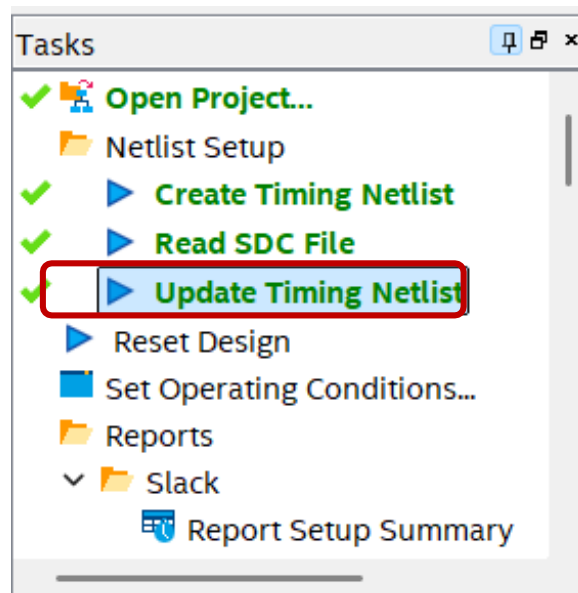
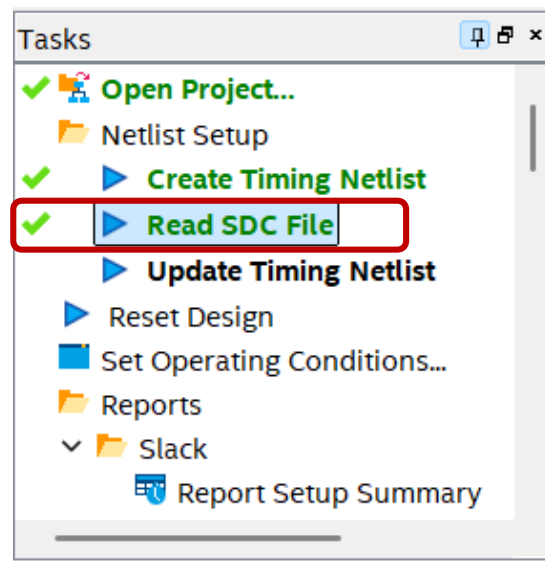
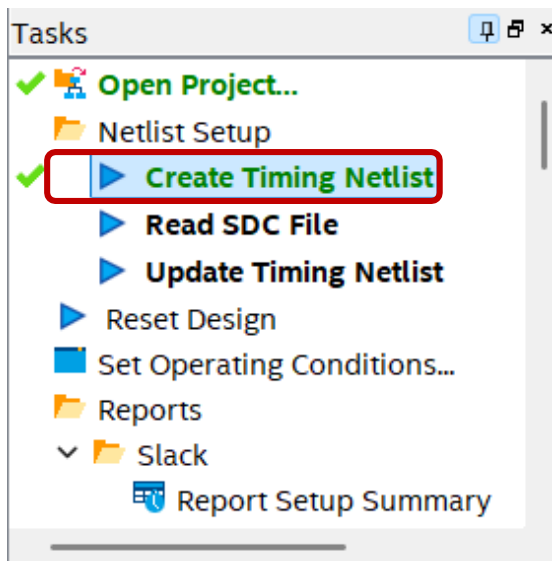
- Under **Timing Analyzer**, browse and add **top.sdc** to the timing constraints file list, then click **Apply** and **OK**.



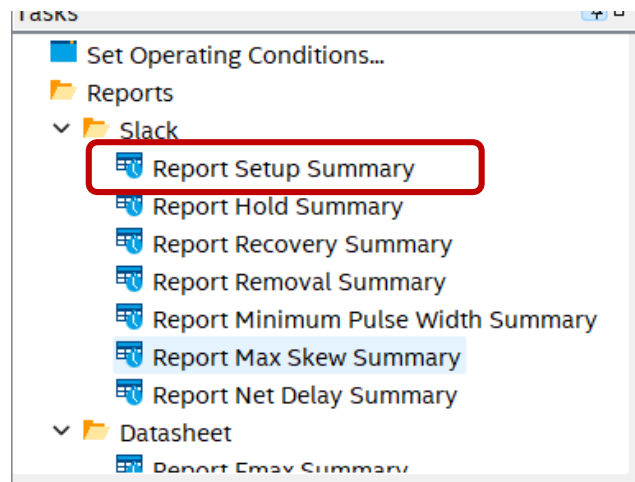
- Launch the tool by selecting **Tools** → **Timing Analyzer** from the top menu bar.



- In the Tasks pane under Netlist Setup, double-click **Create Timing Netlist**, **Read SDC File**, and **Update Timing Netlist**.

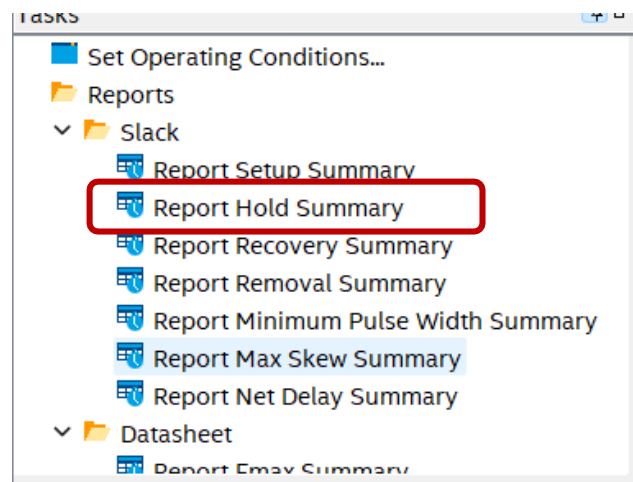


- Expand **Reports** -> **Slack** in the Tasks panel and double-click **Report Setup Summary** to verify a positive setup slack.



Slow 1200mV 85C Model			
	Clock	Slack	End Point TNS
1	u_pll altpll_component auto_generated pll1 clk[0]	97.123	0.000

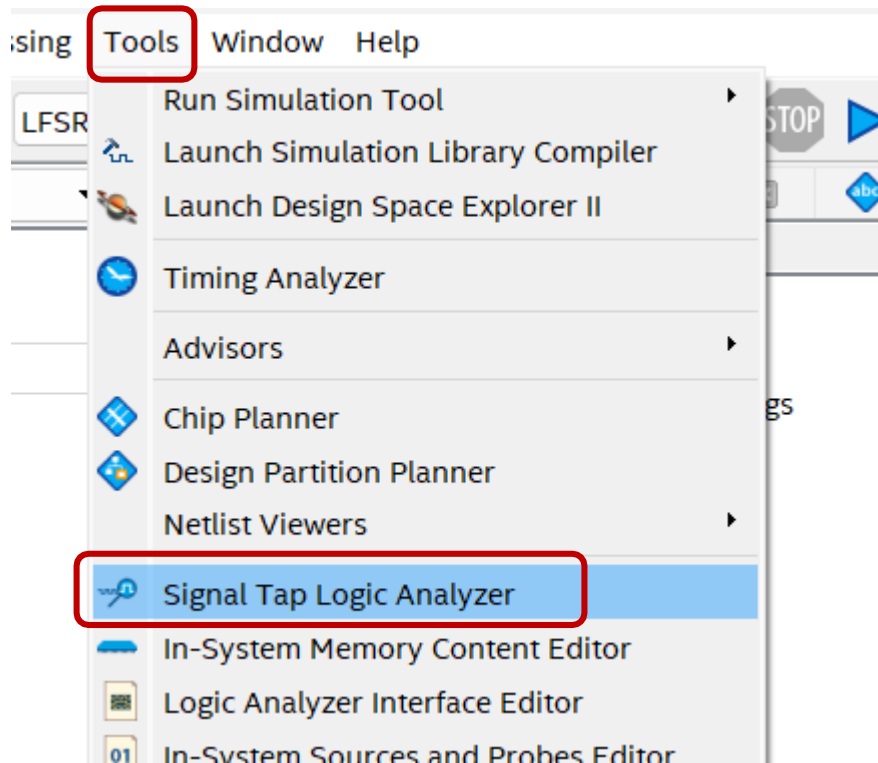
- Double-click **Report Hold Summary** to confirm positive hold slack across all timing corners.



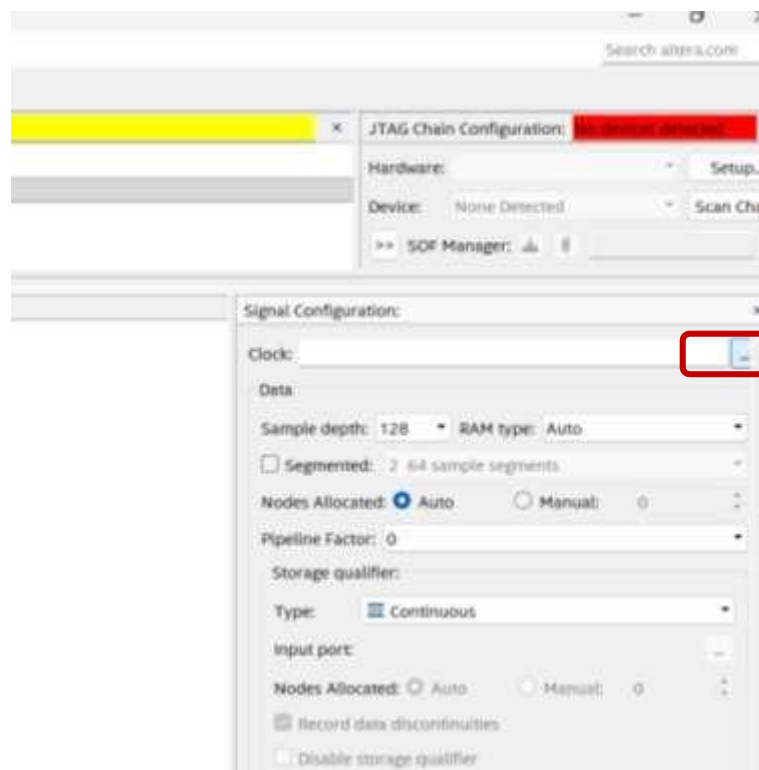
Slow 1200mV 85C Model			
	Clock	Slack	End Point TNS
1	u_pll altpll_component auto_generated pll1 clk[0]	0.454	0.000

Step 5: Logic Analyzer

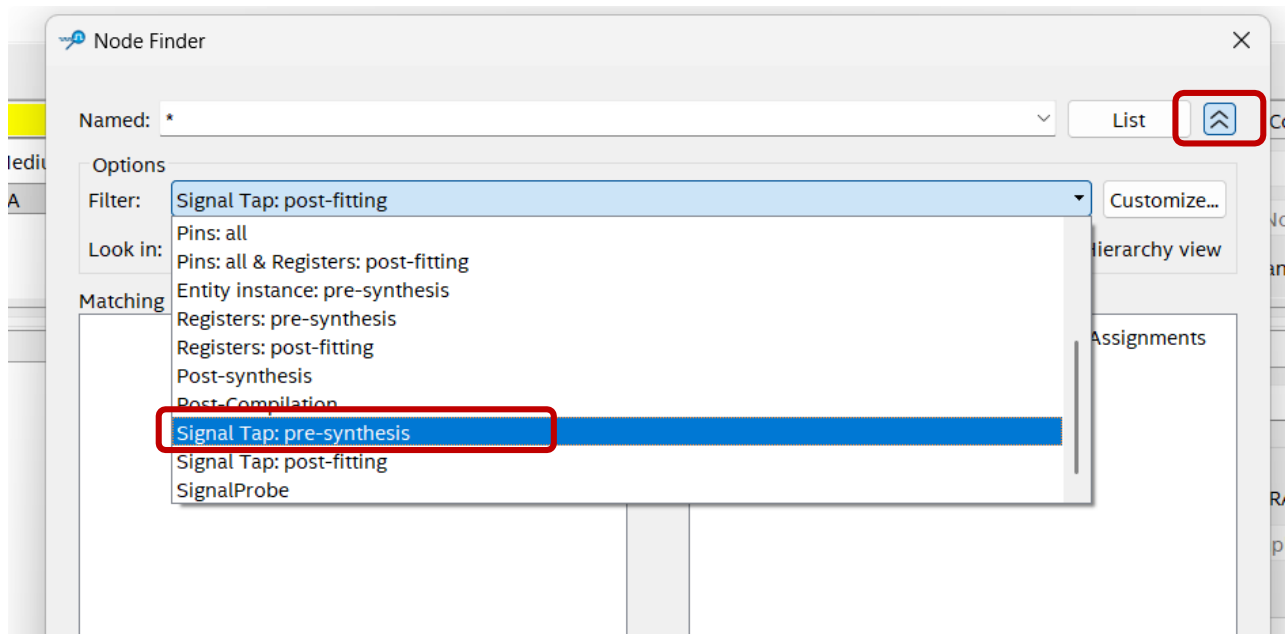
- Launch the tool by selecting **Tools -> Signal Tap Logic Analyzer** from the top menu bar.



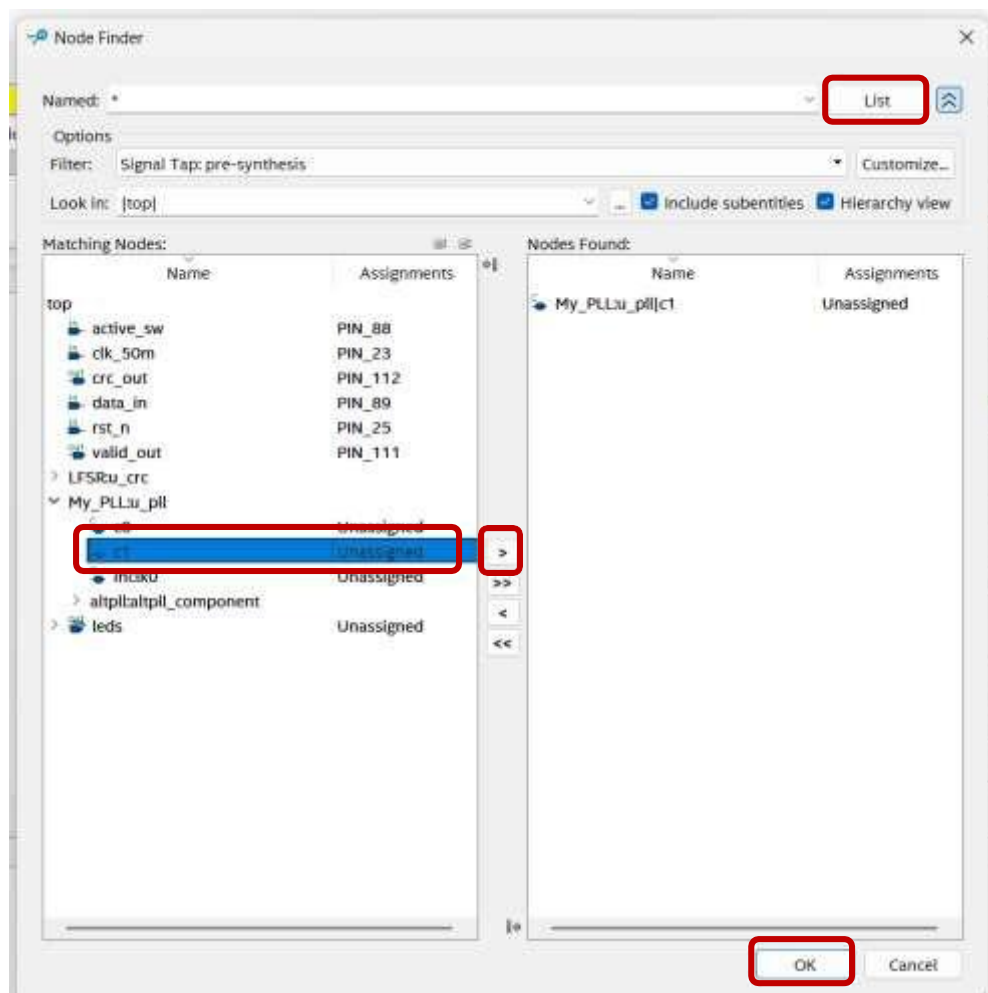
- In the **Signal Configuration** panel on the right, click the ... button next to **Clock** to assign the sampling clock.



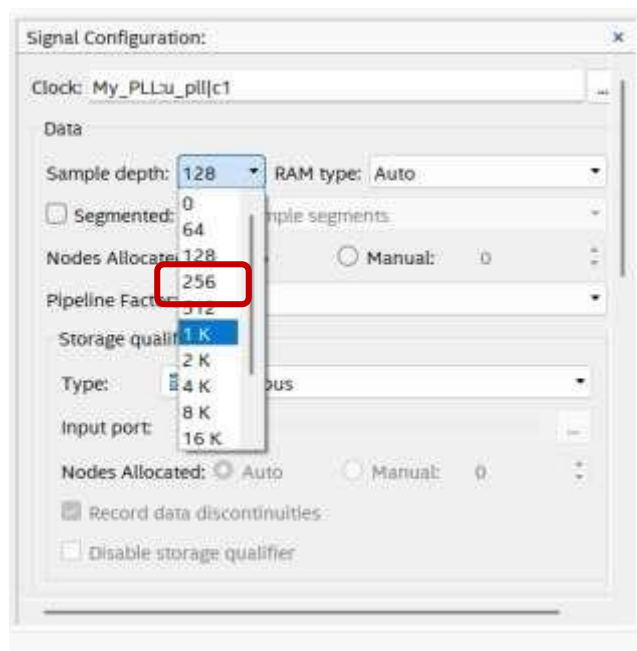
- In the Node Finder window, set the **Filter** to **Signal Tap: pre-synthesis** and click **List**.



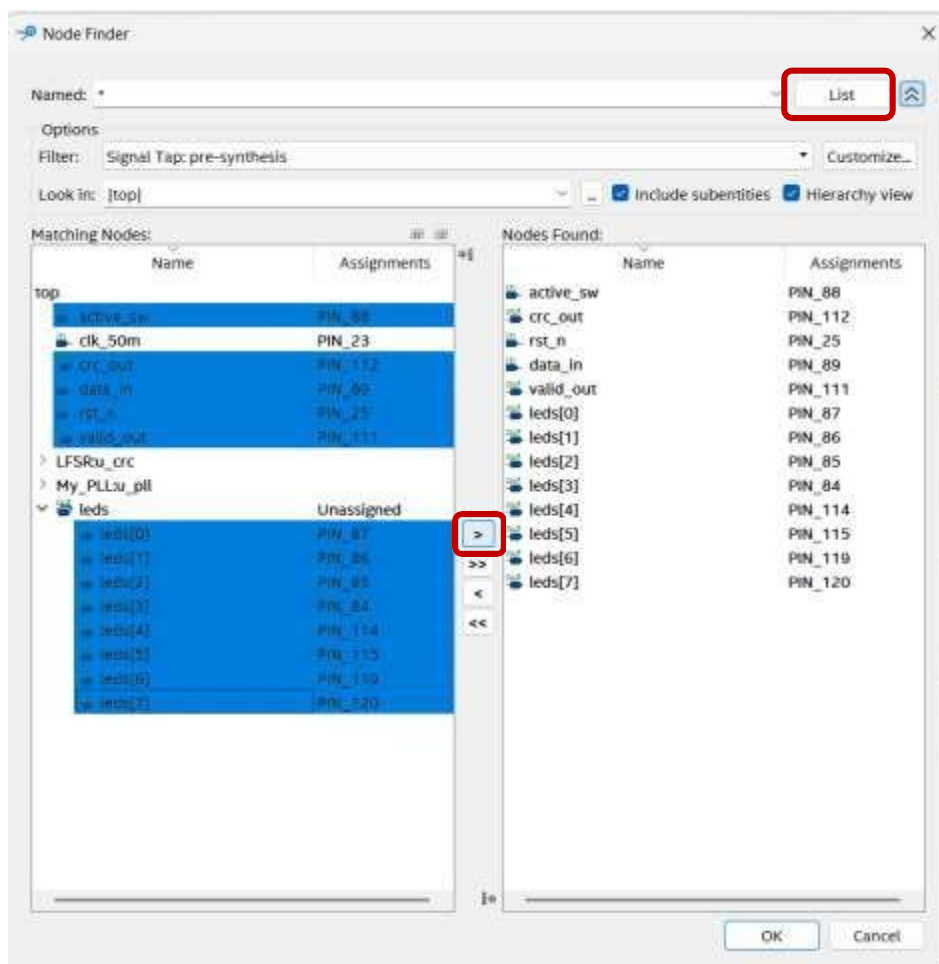
- Select the fast PLL clock output (**My_PLL:u_pll|c1**), click > to add it to **Nodes Found**, and click **OK**.



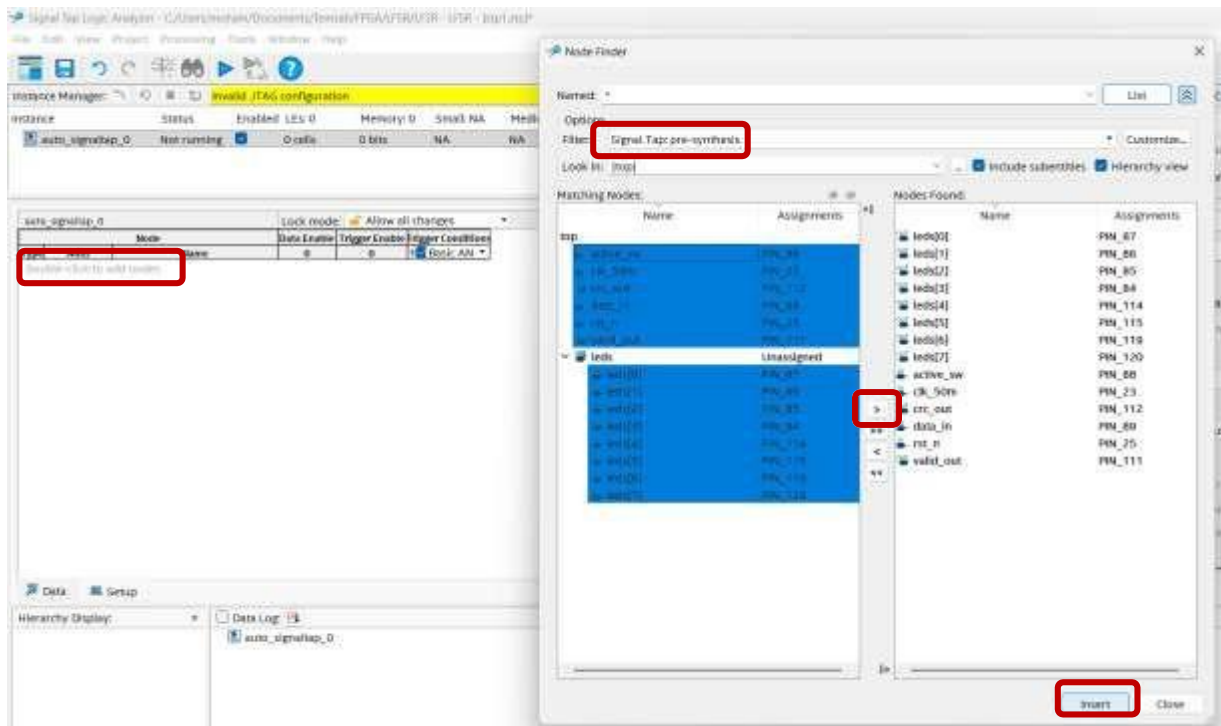
- Change the **Sample depth** setting to **1 K** (1024 samples) in the Signal Configuration pane.



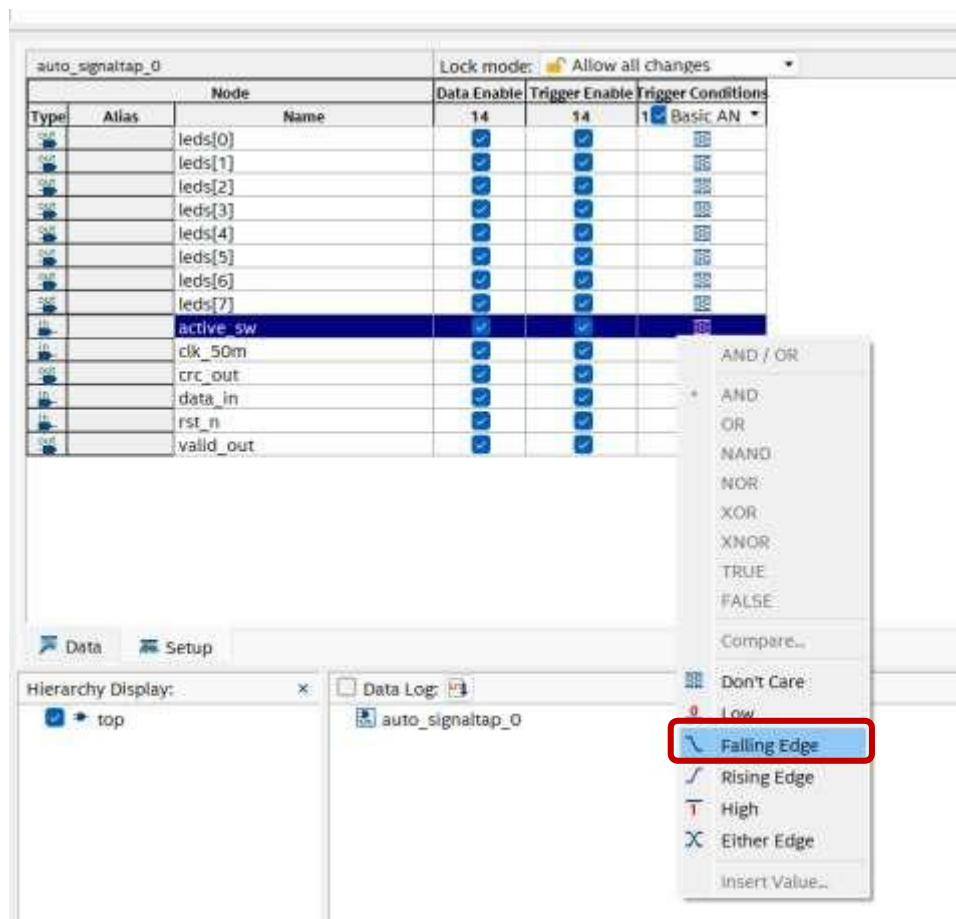
- Double-click the Setup table to open Node Finder, select all project signals (active_sw, crc_out, data_in, rst_n, valid_out, leds[7:0]), and click > to add them.



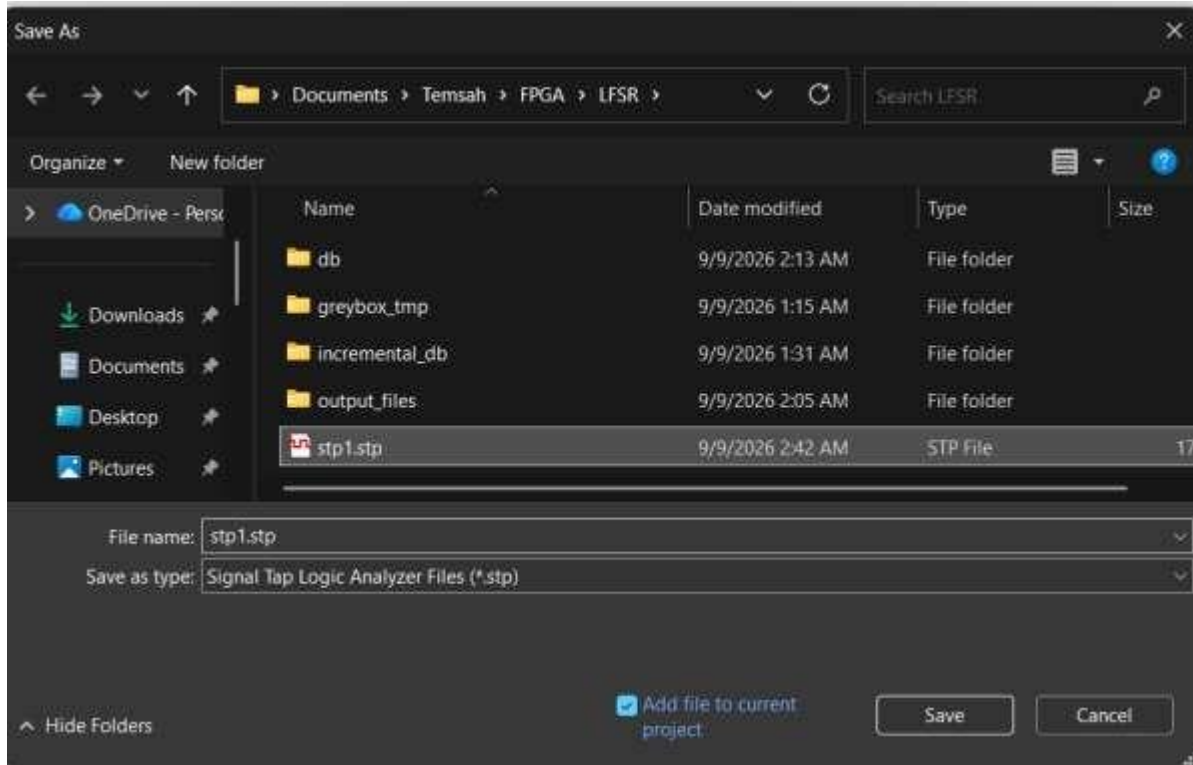
- Click **OK** to insert all 13 signals into the Signal Tap setup grid.



- Right-click the **Trigger Conditions** cell for **active_sw** and select **Falling Edge (\)** to set the trigger.



- Save the file as **stp1.stp** in the project directory and click **Yes** to enable Signal Tap in the active project.



- In the Tasks pane on the left, double-click **Compile Design** to run full synthesis and embed Signal Tap into the hardware bitstream

Tasks			Compilation			
		Task				Time
✓	▼	▶ Compile Design				00:00:48
✓	>	▶ Analysis & Synthesis				00:00:36
✓	>	▶ Fitter (Place & Route)				00:00:06
✓	>	▶ Assembler (Generate programming files)				00:00:02
✓	>	▶ Timing Analysis				00:00:04
	>	▶ EDA Netlist Writer				
		■ Edit Settings				
		■ Program Device (Open Programmer)				

